NYU

ROB-UY 3303

Robot Motion & Planning

Lecture 10A - Expansive Planners



ROB-UY 3303

Robot Motion & Planning

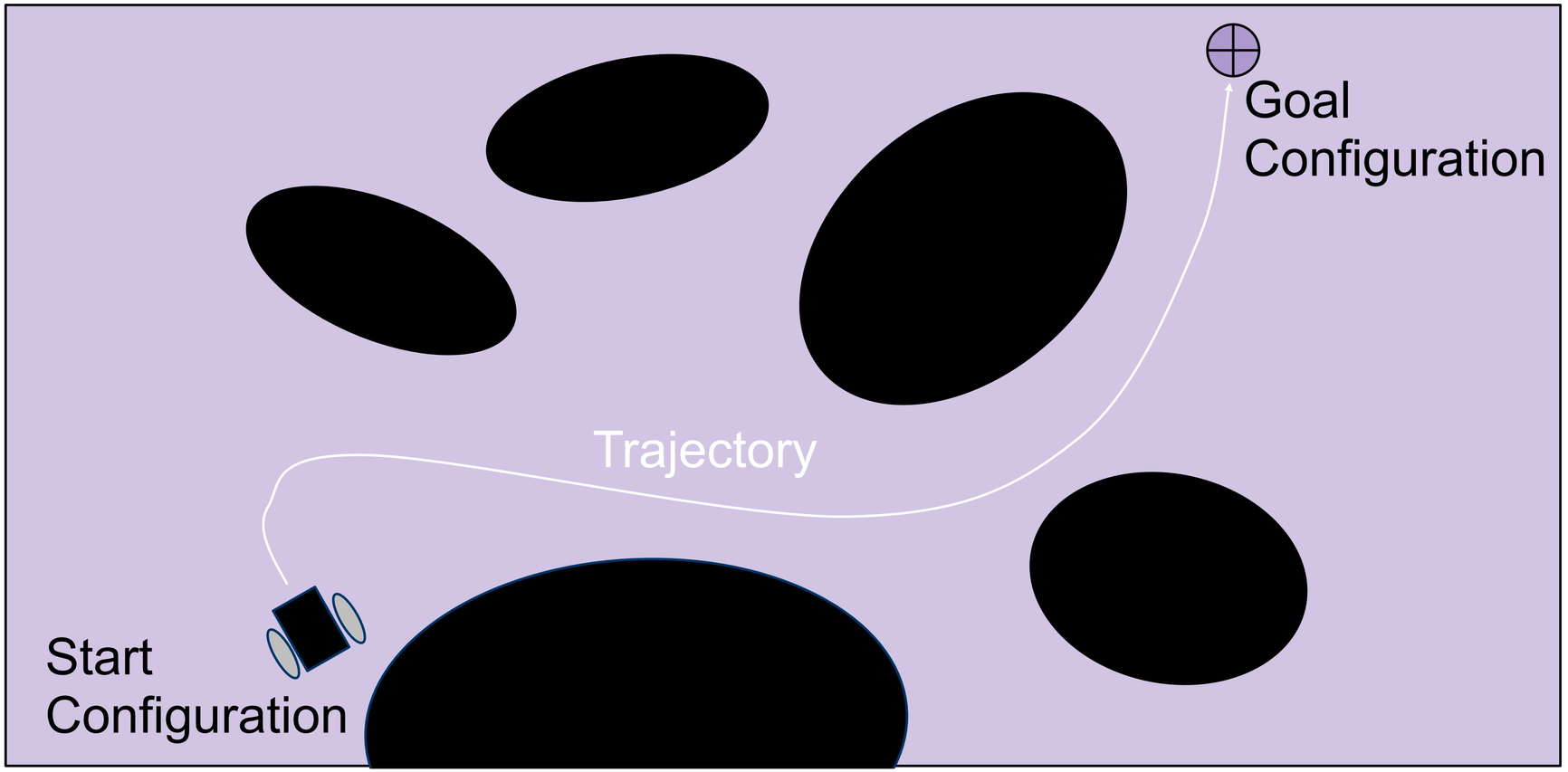


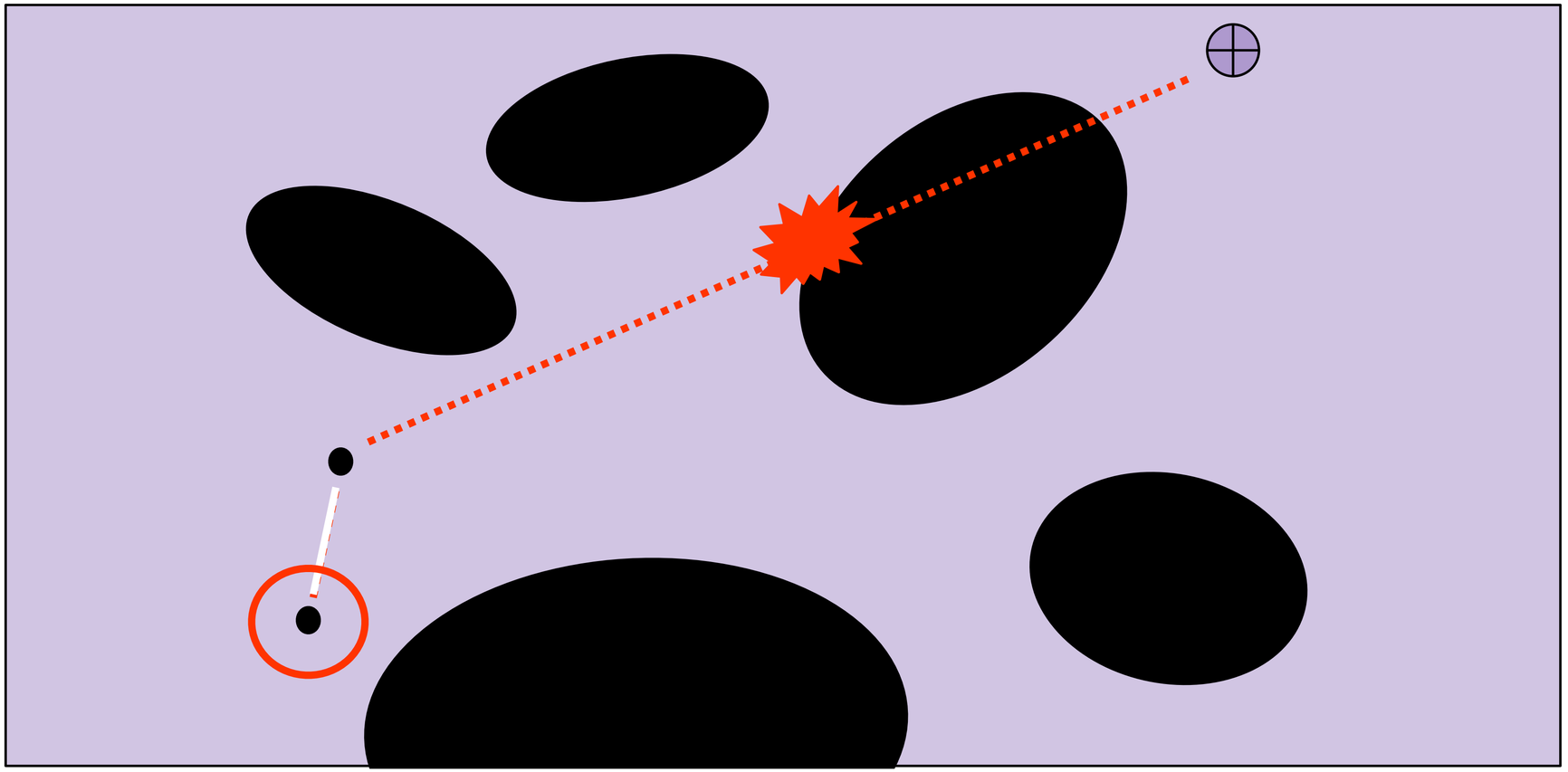
Agenda

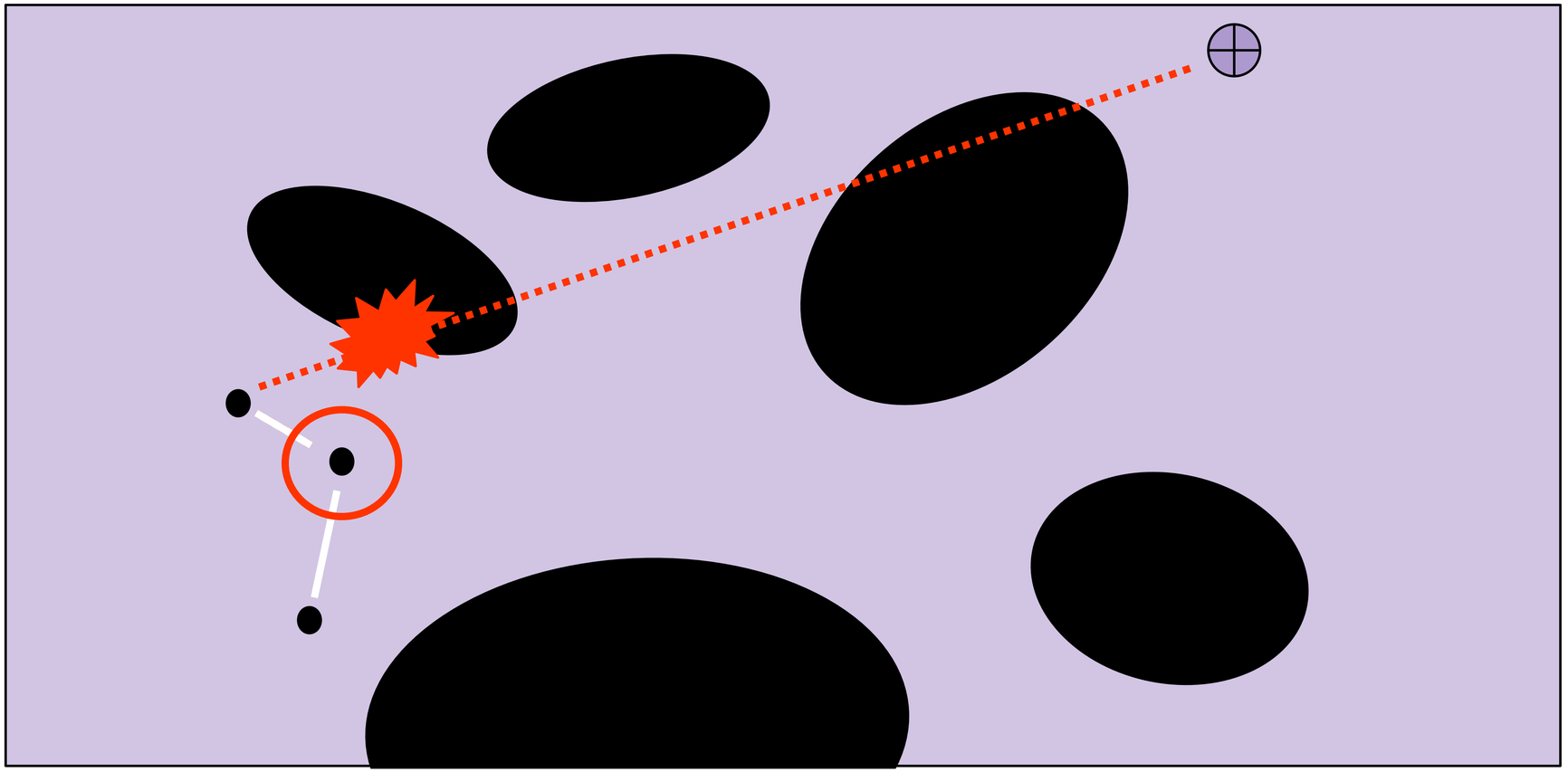
1. Algorithm run-through
2. Pseudo Code
3. Example Implementations

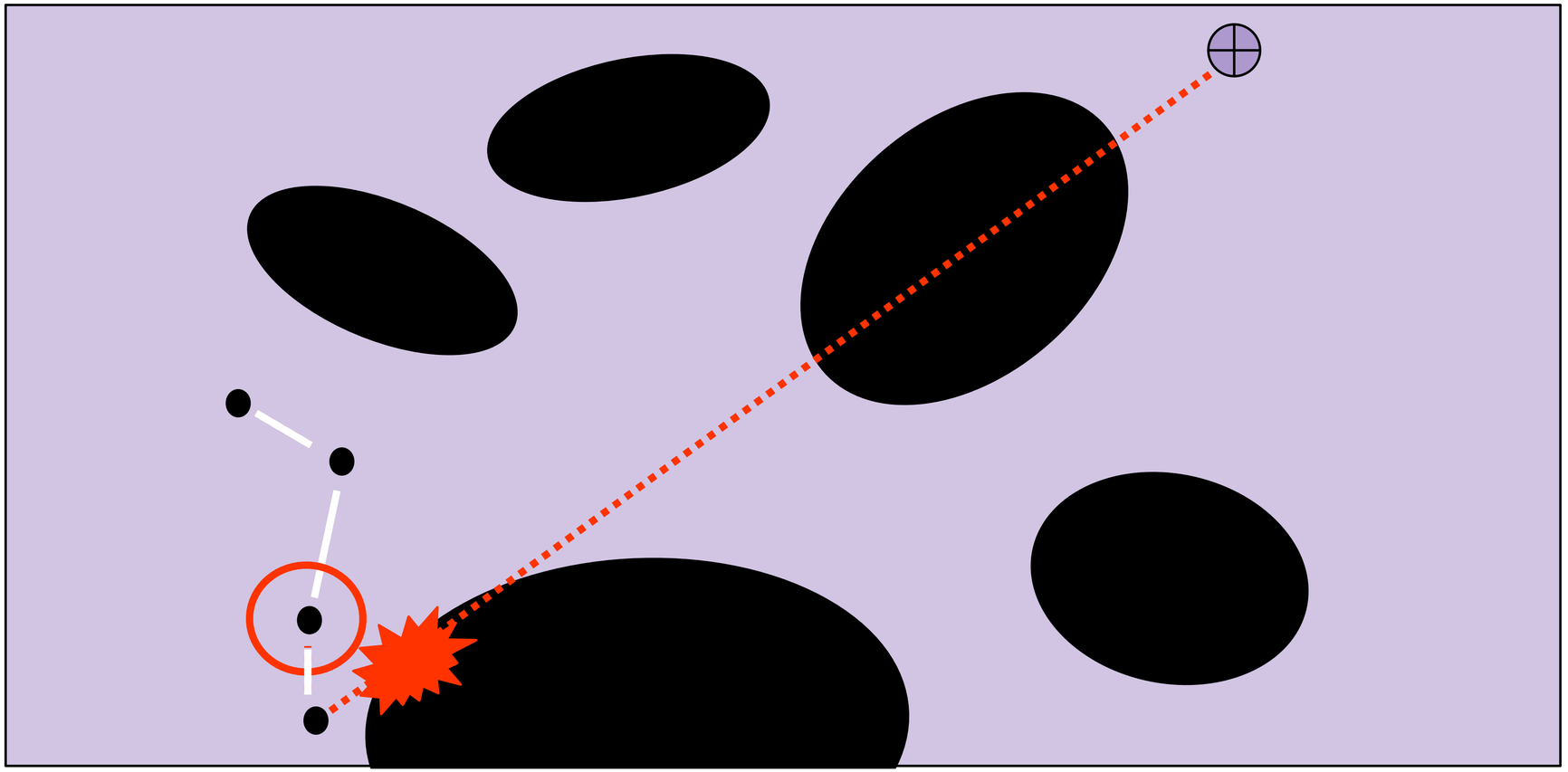
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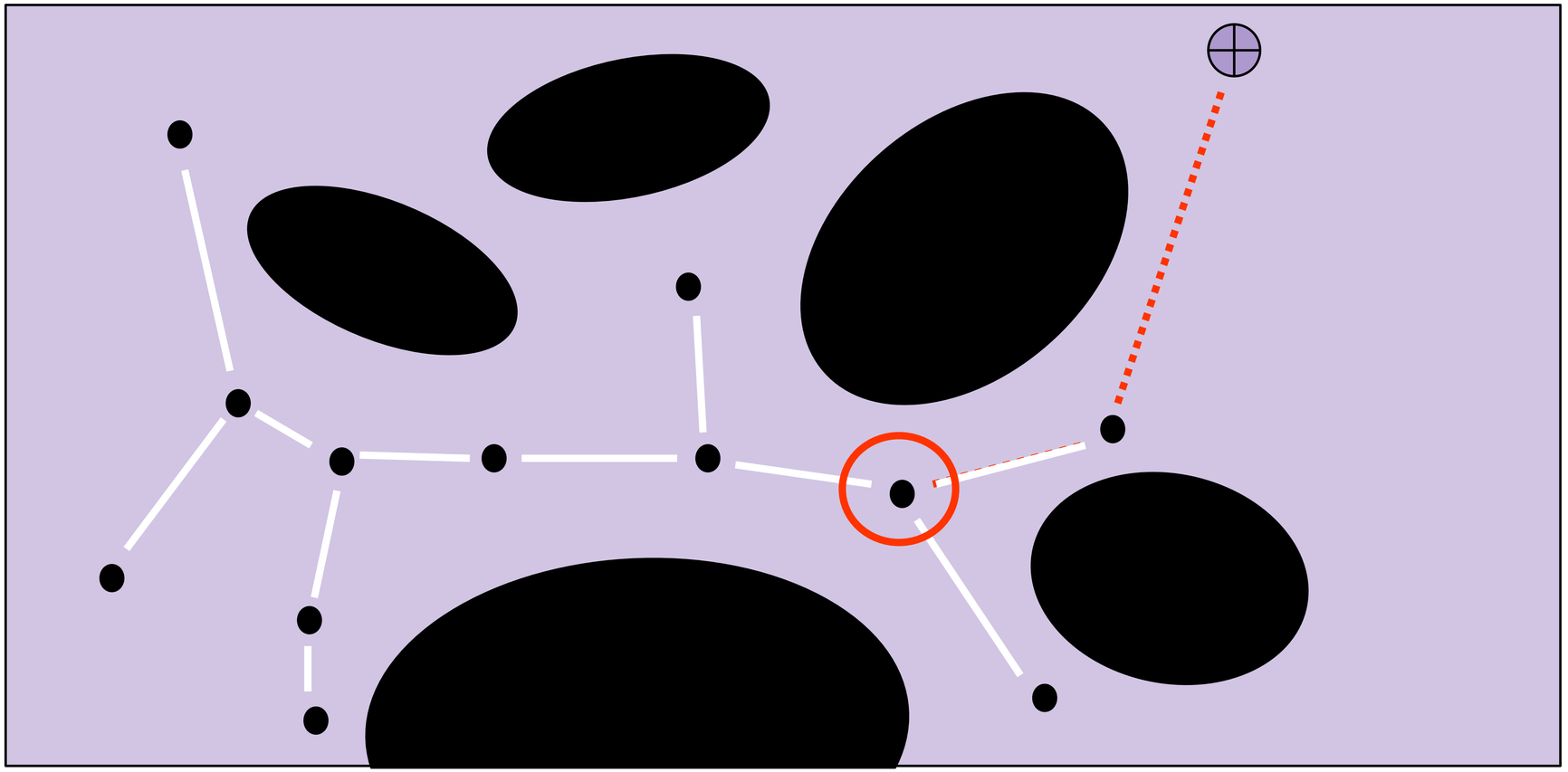
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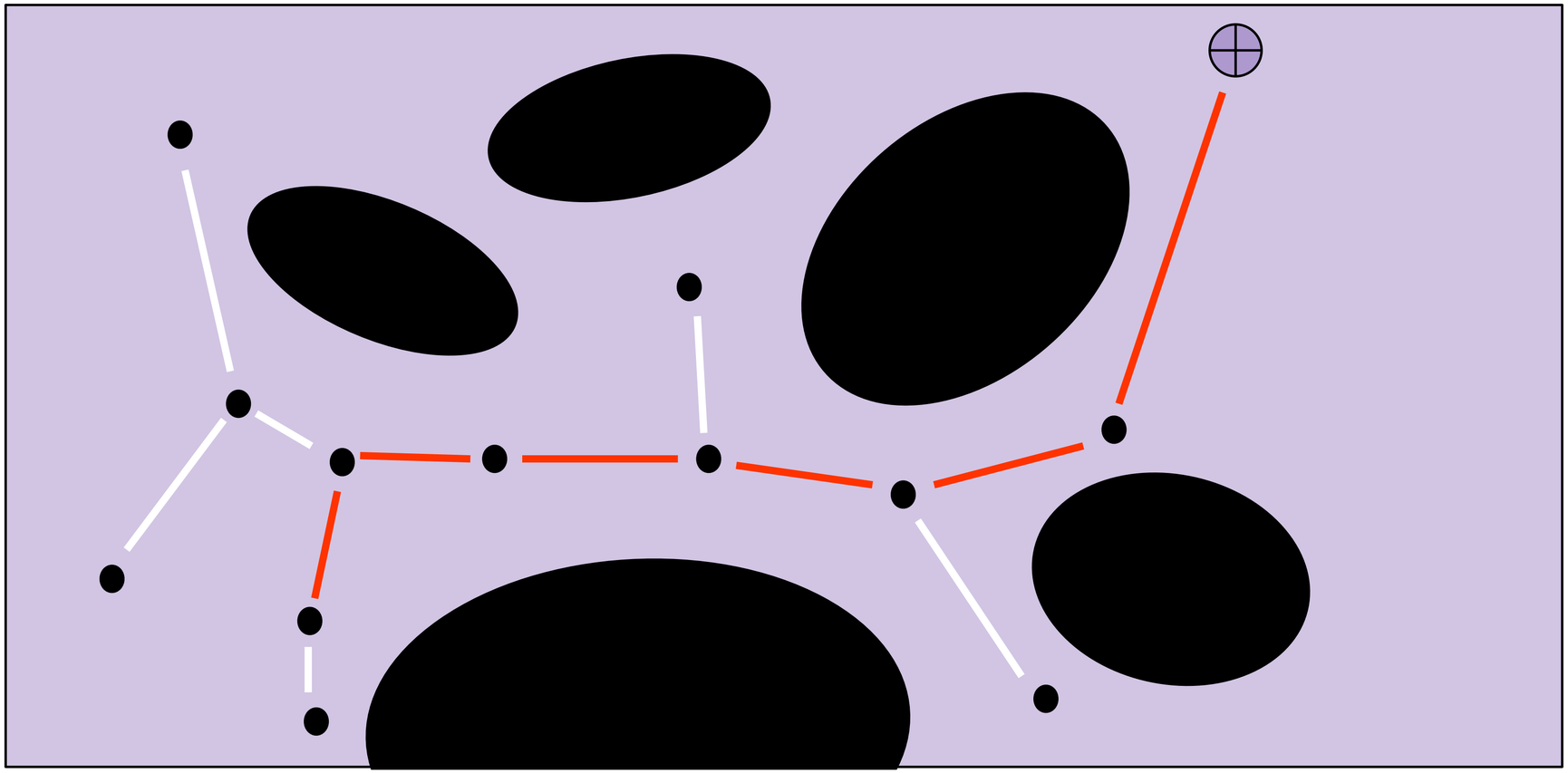

Goal
Configuration
Trajectory
Start
Configuration

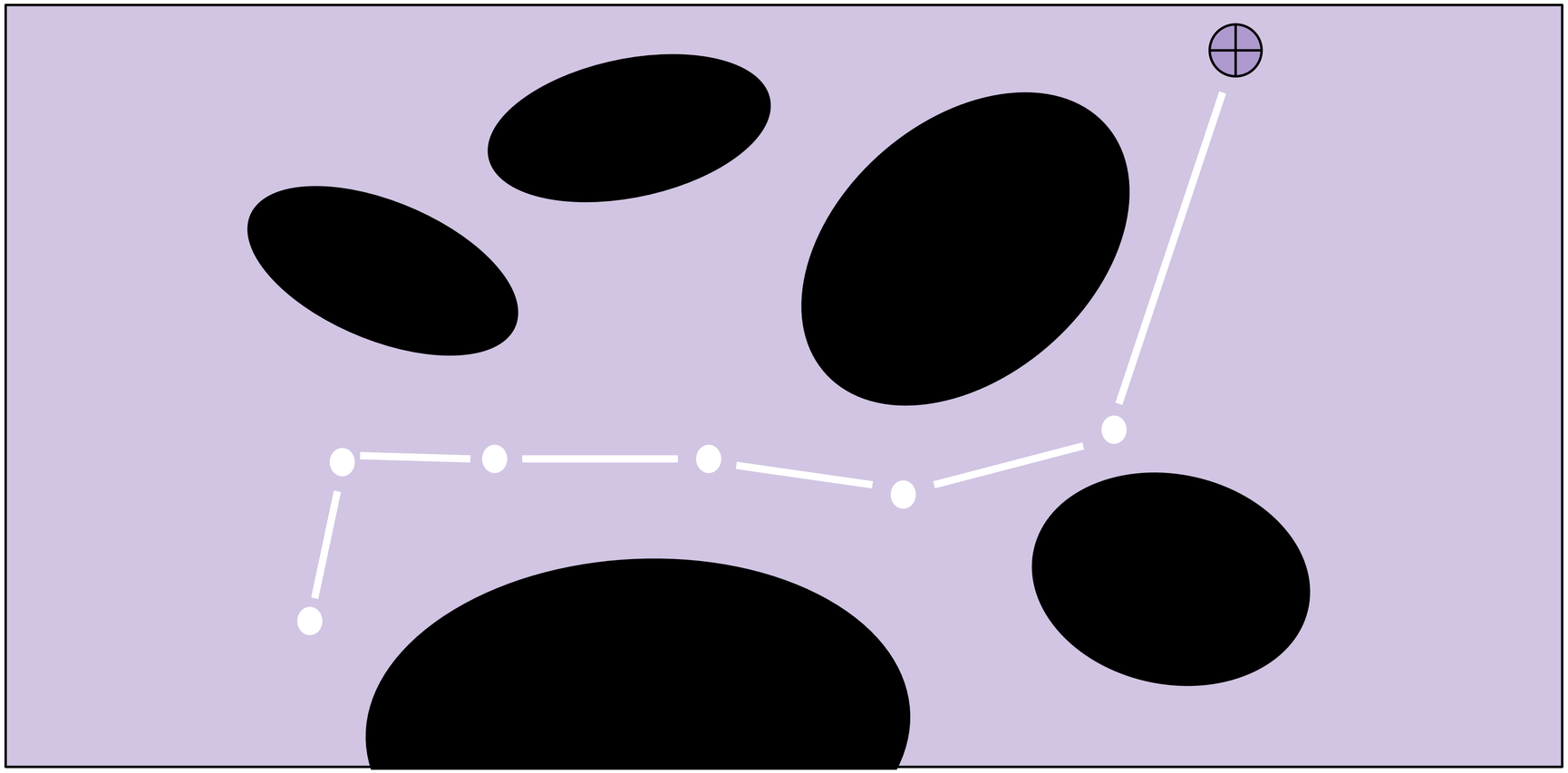












Agenda

1. Algorithm run-through
2. **Pseudo Code**
3. Example Implementations

Pseudo Code

1. add start configuration c_s as node n_s to $R(\mathbf{N}, \mathbf{E})$
2. while not done
3. Randomly select node n from R to expand
4. Randomly Generate new Node n' from n
5. If edge e from n to n' is collision-free
6. Add (n', e) to R
7. If n' belongs to endgame region, return traj

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Three key steps

1. Random node selection from tree (step 3)
2. Random expansion to generate new node (step 4)
3. Endgame region check (step 7)

1. Random Node selection

- **Naive approach:** pick from all nodes in the roadmap with equal probability.
  - Easy to implement
  - Leads to poor expansion

1. Random Node selection

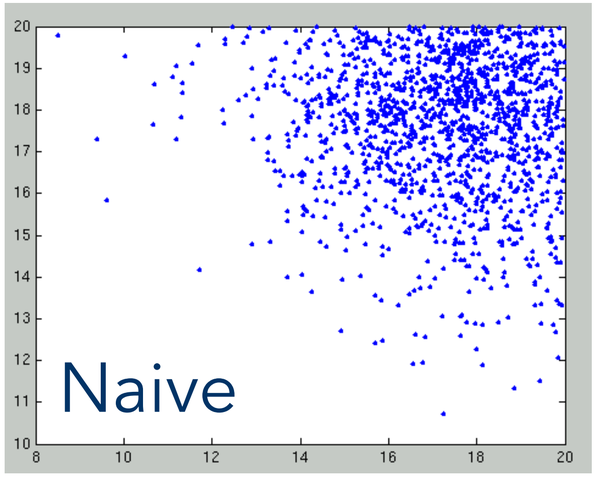
- **Naive approach:** pick from all nodes in the roadmap with equal probability.
  - Easy to implement
  - Leads to poor expansion → **Clustering**

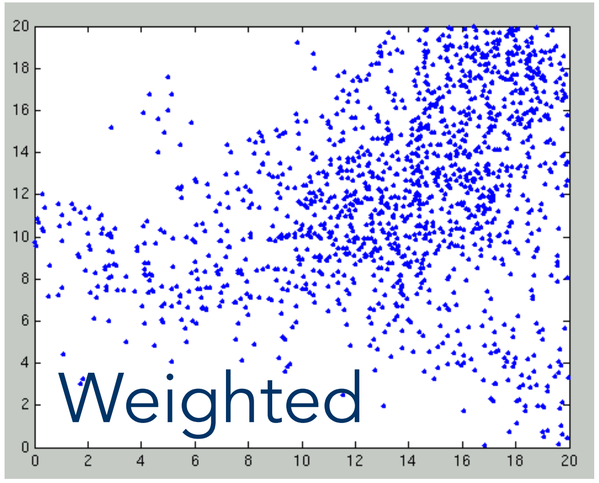
1. Random Node selection

- **Better approach**: pick nodes with probability proportional to the inverse of local node density

1. Random Node selection

- **Example**: A 2DOF configuration space where the initial node in the roadmap is located in the upper right corner.
  - After X iterations, the roadmap produced from an unweighted expansion has limited coverage.

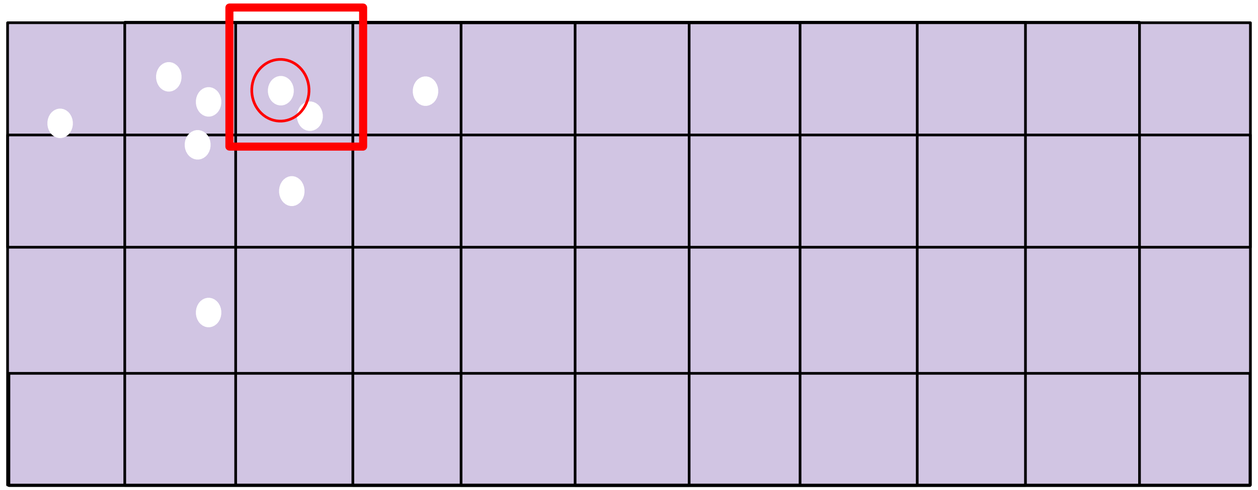




1. Random Node selection

- **Kindel and Hsu's approach**: Divide workspace into grid of cells, then use steps:

1. Randomly pick an occupied cell from the grid.
2. Randomly pick a milestone in that cell.



[Kindel 2000]

2. Random Expansion

- Use random control inputs to propagate robot from previous node c to new configuration c'
- Algorithm:

1. Randomly select controls u and Δt
2. Use known dynamics/kinematics equation f of robot to generate new configuration

$$c' = f(c, u, \Delta t)$$

3. If path from c to c' is collision-free, then add c' to R

2. Random Expansion

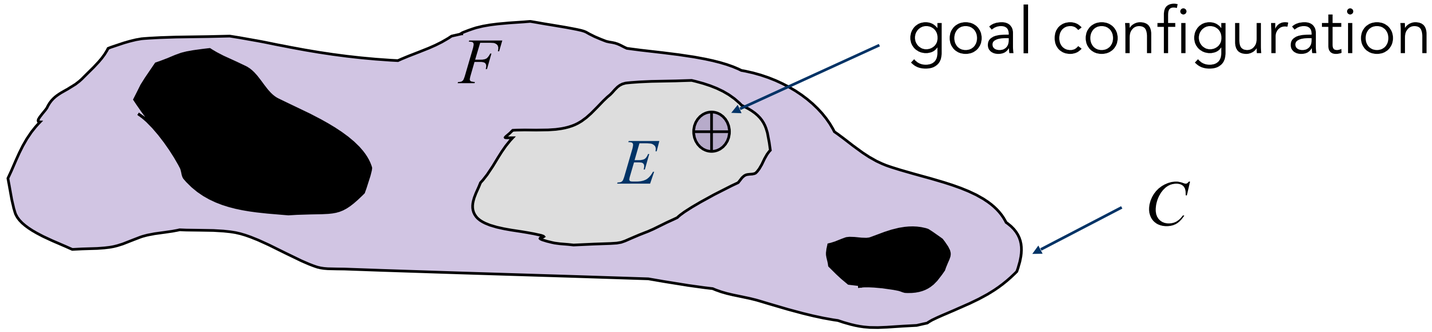
- Example: Differential Drive Robot
- Algorithm:
  1. Randomly select right and left wheel distances Δs_l and Δs_r
  2. Generate new configuration

$$c' = f(x, y, \theta, \Delta s_r, \Delta s_l) = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} + \begin{bmatrix} \frac{\Delta s_r + \Delta s_l}{2} \cos\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r + \Delta s_l}{2} \sin\left(\theta + \frac{\Delta s_r - \Delta s_l}{2b}\right) \\ \frac{\Delta s_r - \Delta s_l}{b} \end{bmatrix}$$

  3. If path from c to c' is collision-free, then add c' to R

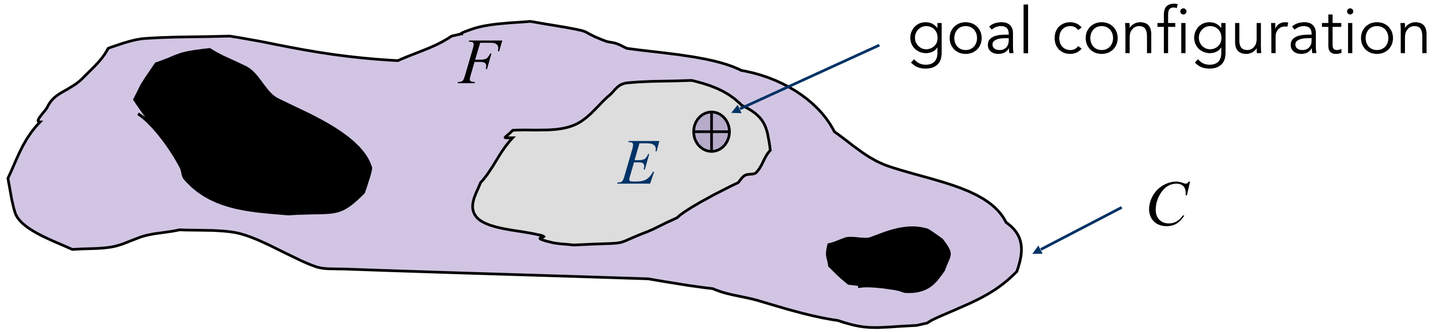
3. Endgame Region Definition

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- For each planning problem, we can define a unique method of making **simple** connections.

3. Endgame Region Definition

- Given the complexity of most configuration spaces, it is very difficult to model E.
- In practice, we develop a simple admissibility test to calculate if a configuration c' belongs to the E
- At every iteration of the algorithm, this test is used to determine if newly generated configurations are connected to the goal configuration.

3. Endgame Region Definition

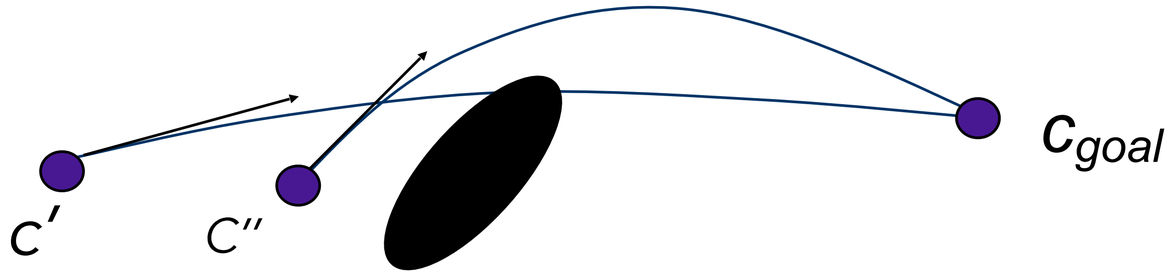
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  1. The set of all configurations within some radius r of the goal configuration
  2. The set of all configurations that have “simple”, collision-free connection with the goal configuration. **Example:** Use circular arc for differential drive robots.



3. Endgame Region Definition

- In defining E, we need two things for good performance:

1. The region E should be **large**: this increases the chance that a newly generated milestone will belong to E and provide us a solution.
2. The admissibility test to be as **fast** as possible. This test is conducted at every iteration of the algorithm and will greatly affect the algorithm running time.

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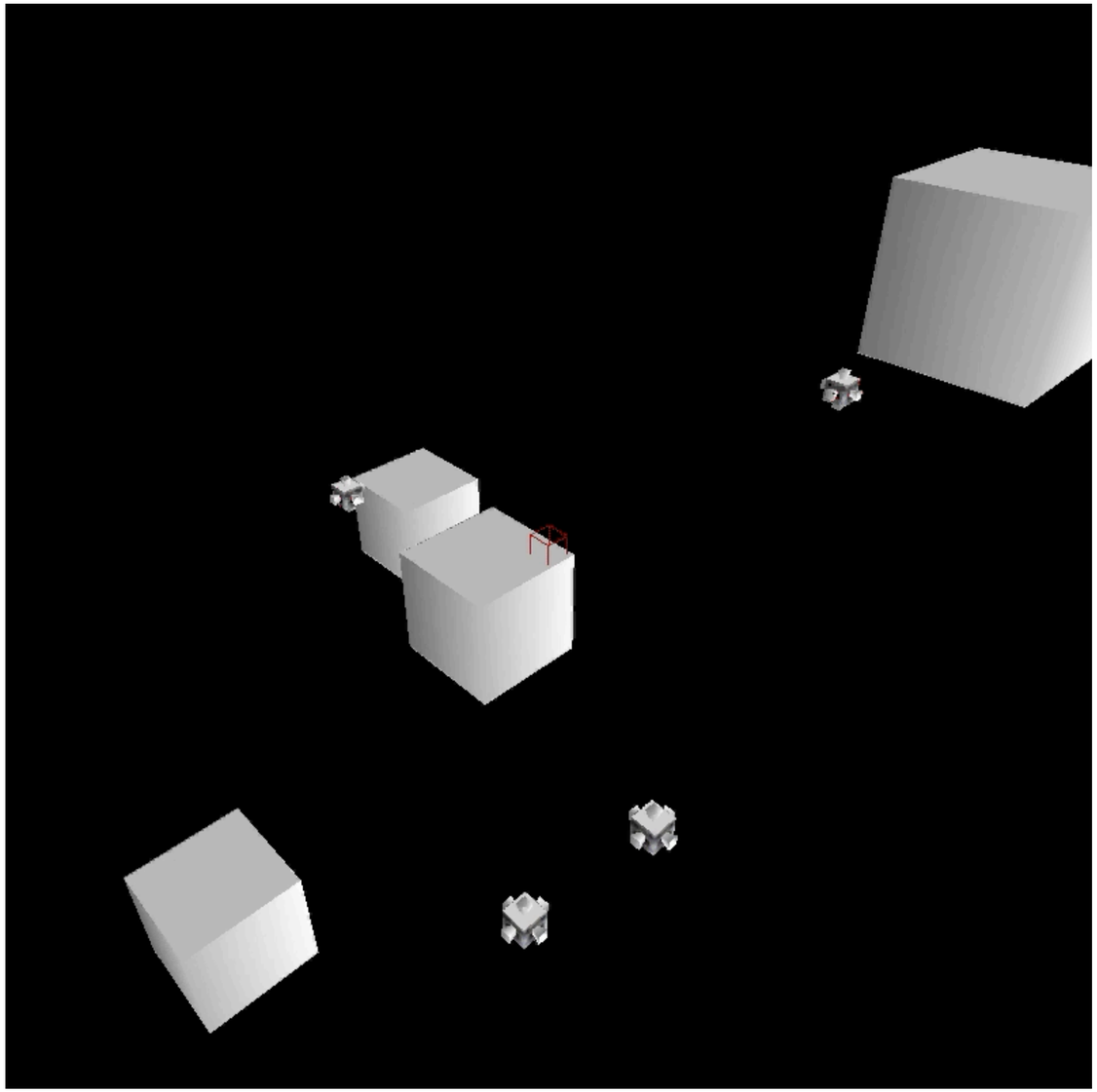




Courtesy of T. Bretl

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